

On the Coalgebraic Linear-Time Semantics of Two-Player Games

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Based on j.w.w. Corina Cîrstea

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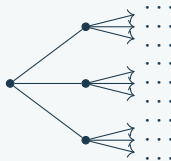
Branching-Time vs Linear-Time

- Linear-time: time is a line.



- One reasons about all execution paths of a program or system¹.
- $GF\varphi$ means “ φ always holds at the some point in the future”.

- Branching-time: time is a tree.



- $\forall G \exists F \varphi$ means “at every future point, there exists a branch along which we eventually encounter φ ”
- The linear-time paradigm is typically preferred in *reactive synthesis*².
 - $\psi \subseteq I \times O$, does $\forall i \exists o : \psi(i, o)$?
 - *Skolemise*: $\exists f : I \rightarrow O, \forall i \in I : \psi(i, f(i))$
 - f viewed as a *strategy* in a *two-player game*.

¹Advocated in “Sometime” Is Sometimes “Not Never” [Lamport, 1980].

²On the Synthesis of a Reactive Module [Pnueli and Rosner, 1989].

Motivation: Model Checking & Synthesis

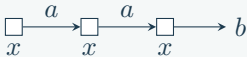
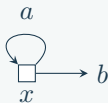
- Model checking is about how a system X behaves.
 - Does X satisfy P ?
- Synthesis is about what behaviour a system X can enforce.
 - Can X enforce P ?
- *Coalgebra* studies morphisms $\gamma : X \rightarrow F(X)$
 - Generates F -behaviour for some $F : \mathbb{C} \rightarrow \mathbb{C}$
 - Coalgebra has found several applications in model checking³⁴⁵.
- What about synthesis?
 - Behaviour $\mapsto$ *enforceable* behaviour

³A Unifying Approach to Product Constructions for Quantitative Temporal Inference [Watanabe et al., 2025]

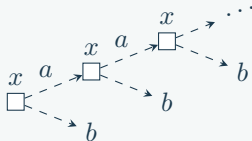
⁴Lattice-Theoretic Progress Measures and Coalgebraic Model Checking [Hasuo et al., 2016]

⁵Measure-Theoretic Semantics for Quantitative Parity Automata [Cirstea and Kupke, 2023]

Labelled transition systems:



Labelled games:



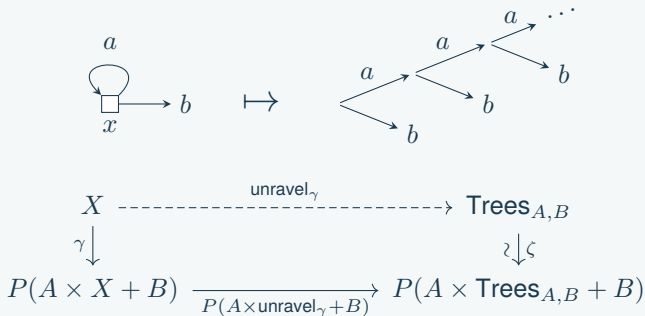
Branching-Time Semantics

$X \mapsto P(A \times X + B)$ is a functor **Set** $\rightarrow$ **Set**.

Labelled transition systems with termination⁶:

$$\gamma : X \rightarrow P(A \times X + B)$$

(X, γ) has a canonical branching-time semantics $\text{unravel}_\gamma : X \rightarrow \text{Trees}_{A,B}$:

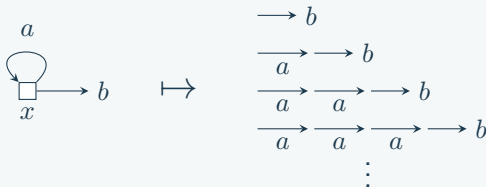


⁶These are Milner's *charts* [Milner, 1984].

Finite Linear-Time Semantics

$$\frac{X \rightarrow P(A \times X + B) \text{ in } \mathbf{Set}}{X \mapsto A \times X + B \text{ in } \mathbf{Rel} \cong \mathbf{Kl}(P)}$$

(X, γ) has a canonical *finite* linear-time semantics⁷ $\text{ftrace}_\gamma : X \mapsto A^*B$:



$$\begin{array}{ccc} X & \xrightarrow{\text{ftrace}_\gamma} & A^*B \\ \gamma \downarrow & & \wr \downarrow \iota^{-1} \\ A \times X + B & \xrightarrow{A \times \text{ftrace}_\gamma + B} & A \times A^*B + B \end{array}$$

⁷This is an instance of *Generic Trace Semantics via Coinduction* [Hasuo et al., 2007] parameterised by a monad $T := P$ and a functor $F := A \times (-) + B$.

Infinite Linear-Time Semantics

$$\begin{array}{ccc}
 X & \xrightarrow{f} & A^\omega + A^*B \\
 \gamma \downarrow & & \wr \downarrow \zeta \\
 A \times X + B & \xrightarrow{A \times f + B} & A \times (A^\omega + A^*B) + B
 \end{array}$$

❶ Uniqueness fails⁸:



$$x \mapsto \emptyset$$

$$x \mapsto \mathbf{GF}a_1$$

$$x \mapsto \mathbf{FG}a_1$$

$$x \mapsto \{a_1, a_2\}^\omega$$

❷ $\mathbf{Rel}(X, A^\omega + A^*B)$ is a complete lattice

❸ $f \mapsto^{\Phi} \zeta^{-1} \odot A \times f + B \odot \gamma$ is a monotone function

❹ There is a greatest and least homomorphism (or fixed point)

- The least homomorphism only records finite traces
- The greatest trace $_{\gamma} : X \rightarrow A^\omega + A^*B$ records all traces

⁸This is starting point of *Trace Semantics for Coalgebras* [Jacobs, 2004]

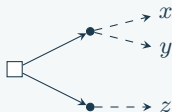
Outline

- 1 Identify *monads* to model the branching structure of games.
 - The controller $\square$ will be nondeterministic.
 - The environment $\bullet$ will be nondeterministic/probabilistic/weighted.
- 2 Obtain a finite linear-time semantics map for games.
 - Describe in terms of *finitely completing controller strategies*.
- 3 The greatest homomorphism approach to infinite trace semantics does not generalise.
 - Discuss a definition in terms of categorical limits.
- 4 Conclusion.

Monads for Games (I)

- $PP : \mathbf{Set} \rightarrow \mathbf{Set}$ replaces the powerset functor $P : \mathbf{Set} \rightarrow \mathbf{Set}$.

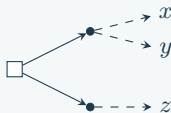
- $\{\{x, y\}, \{z\}\}$ models a game:



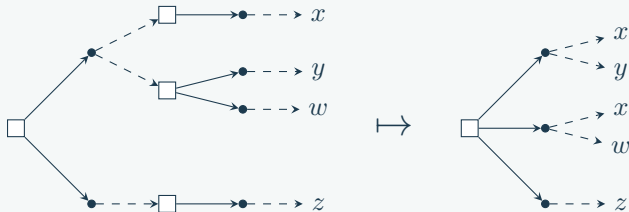
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- $PP : \mathbf{Set} \rightarrow \mathbf{Set}$ replaces the powerset functor $P : \mathbf{Set} \rightarrow \mathbf{Set}$.

- $\{\{x, y\}, \{z\}\}$ models a game:



- How should we define $\mu_X^{PP} : PPPP(X) \rightarrow PP(X)$?



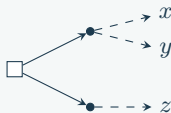
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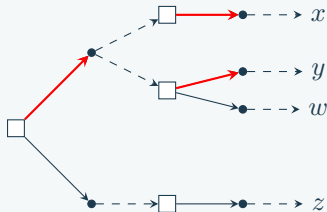
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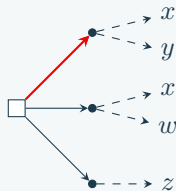
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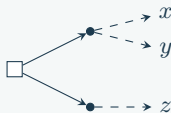
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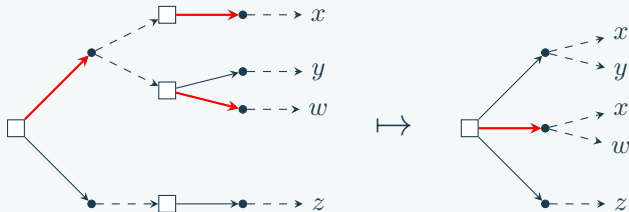
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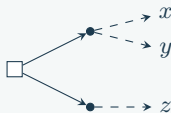
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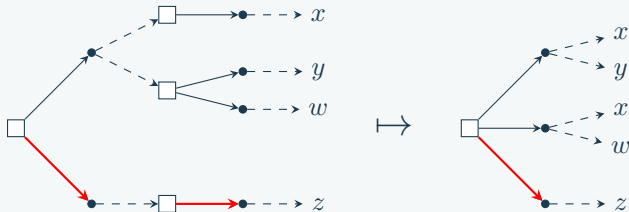
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Monads for Games (II)

- μ^{PP} fails to be associative⁹¹⁰
 - It is associative precisely when we impose union-closure¹¹.
 - $\widetilde{PP}(X) := \{\mathcal{U} \in PP(X) \mid \mathcal{U} \text{ is closed under non-empty union}\}$.
 - μ^{PP} preserves the property of *upwards-closure*
 - $\mathcal{M}(X) := \{\mathcal{U} \in PP(X) \mid \mathcal{U} \text{ is upwards closed}\}$

⁹Contradicting *Monads of Sets* [Manes, 2003]

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- The monad $\widetilde{PP} : \mathbf{Set} \rightarrow \mathbf{Set}$ arises from a *weak* distributive law:
 - A natural transform $\delta : PP \rightarrow PP$:



- $V \in \delta_X(\mathcal{U})$ iff V “hits” every subset in $\mathcal{U}$ and “misses” $X \setminus \bigcup \mathcal{U}$
- Algebraically: two complete lattices $(X, \vee)$ and $(X, \wedge)$ with complete distributivity.

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 - Algebraically: two complete lattices $(X, \vee)$ and $(X, \wedge)$ with complete distributivity.
- The monad $\mathcal{M} : \mathbf{Set} \rightarrow \mathbf{Set}$ arises from a distributive law of the upsets monad over the downsets monad, on $\mathbf{Pos}$, restricted to $\mathbf{Set}$.
 - Algebraically: two complete lattices with $(X, \vee)$ and $(X, \wedge)$ with complete distributivity and absorption.

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Monads for Games (III): Quantitative Opponents

- The generalised distribution monad M_S for a semiring $(S, +, 0, \cdot, 1)$.
 - $M_S(X)$ is the set of finitely supported functions $\varphi : X \rightarrow S$
 - The multiplication $\mu_X^{M_S} : M_S M_S(X) \rightarrow M_S(X)$ take a weighted average:

$$\Phi \in M_S M_S(X) \mapsto \lambda x. \sum_{\varphi \in M_S(X)} \varphi(x)$$

- We have *affine*¹² and *weakly affine*¹³ variants:

$$M_S^{\text{aff}}(X) := \{\varphi \in M_S(X) \mid \sum_{x \in \text{supp } \varphi} \varphi(x) = 1\}$$

$$M_S^{\text{weak}}(X) := \{\varphi \in M_S(X) \mid \sum_{x \in \text{supp } \varphi} \varphi(x) \neq 0\}$$

• Examples

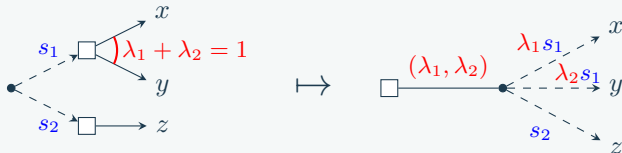
- $(\mathbb{N}, +, 0, \cdot, 1)$. $M_{\mathbb{N}}$ is the *multiset monad*
- $(\mathbb{R}_+, +, 0, \cdot, 1)$. $M_{\mathbb{R}_+}^{\text{aff}}$ is the *finite distribution monad*
- $(\mathbb{B}, \vee, \perp, \wedge, \top)$. $M_{\mathbb{B}}$ is the *finite powerset monad*
- $(\mathbb{R}^\infty, \min, \infty, +, 0)$ models costs/energy

¹²See *Affine parts of monad* [Lindner, 1979].

¹³See *Weakly Markov Categories and Weakly Affine Monads* [Fritz et al., 2023].

Monads for Games (IV)

- When do we have a weak distributive law $M_S P \rightarrow P M_S$? When¹⁴:
 - S is *positive*: $x + y = 0 \implies x = y = 0$
 - S is a *semifield*: $x \neq 0 \implies \exists y \in S : y \cdot x = x \cdot y = 1$
 - Examples: $(\mathbb{R}_+, +, \cdot)$, $(\mathbb{B}, \vee, \wedge)$, $(\mathbb{R}^\infty, \min, +)$
 - $(\mathbb{N}, +, \cdot)$ is not a semifield.
- The action of $M_S P \rightarrow P M_S$ resembles that of $PP \rightarrow PP$:



- We obtain a *convex set of generalised distributions monad* $\widetilde{PM}_S(X)$.

¹⁴See *The Monads of Classical Algebra are Seldom Weakly Cartesian* [Clementino et al., 2014] for precise conditions. See also *Convexity via Weak Distributive Laws* [Bonchi and Santamaria, 2022].

Outline

- ➊ Identify *monads* to model the branching structure of games.
 - The controller $\square$ will be nondeterministic.
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- ➋ **Obtain a finite linear-time semantics map for games.**
 - Describe in terms of *finitely completing controller strategies*
- ➌ The greatest homomorphism approach to infinite trace semantics does not generalise.
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- ➍ Conclusion.

Finite Trace Theorem [Hasuo et al., 2007]

Theorem 1

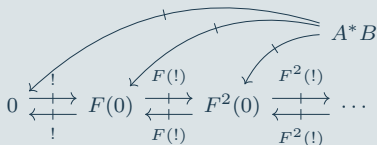
Let T be a monad on **Set**. If

- 1 $T(0) \cong 1$
- 2 $\mathbf{Kl}(T)$ is ω -**cpo**-enriched

Then A^*B is the terminal $A \times (-) + B$ -coalgebra over $\mathbf{Kl}(T)$.

Sketch.

Let $F = A \times (-) + B : \mathbf{Kl}(T) \rightarrow \mathbf{Kl}(T)$.



□

Failure of Enrichment

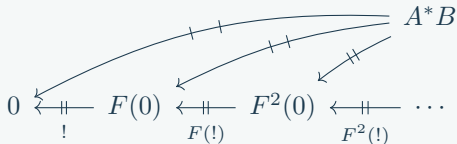
- Monads $\widetilde{PP}$ and $\mathcal{M}$ fail to be enriched in $\omega\text{-}\mathbf{cpo}$, because the environment has infinite branching, let $X = \mathbb{N}$:

	$x = 0$	$x = 1$	$x = 2$	$\dots$
g_0	A	A	A	
g_1	A, B	A	A	
g_2	A, B	A, B	A	
g_3	A, B	A, B	A, B	
$\vdots$				
$\bigvee_n g_n$	A, B	A, B	A, B	

- $\widetilde{PM}_S$ does not suffer from this defect.
 - $\widetilde{PM}_S(0) = \{\emptyset, \{\epsilon\}\}$
 - $\widetilde{PM}_S^c(0) = \{\emptyset\}$ for $c \in \{\text{aff}, \text{weak}\}$

The Finite Trace Map for Games

In $\mathbf{Kl}(\widetilde{PM}_S^c)$:



A^*B is terminal $A \times (-) + B$ -coalgebra over $\mathbf{Kl}(\widetilde{PM}_S^c)$:

$$\begin{array}{ccc}
 X & \xrightarrow{\text{ftrace}_\gamma} & A^*B \\
 \gamma \downarrow & & \wr \downarrow \iota^{-1} \\
 A \times X + B & \xrightarrow{A \times \text{ftrace}_\gamma + B} & A \times A^*B + B
 \end{array}$$

We can show that $\text{ftrace}_\gamma : X \rightarrow \widetilde{PM}_S^c(A^*B)$ arises from the *finitely completing strategies*.

Strategies, categorically.

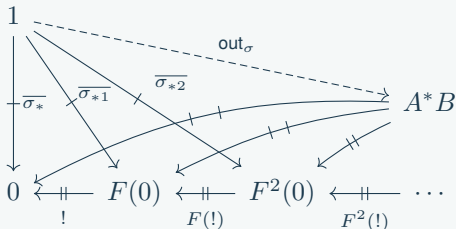
- We have a functor $\overline{(-)} : \mathbf{Kl}(M_S^c) \rightarrow \mathbf{Kl}(\widetilde{PM}_S^c)$
- In $\mathbf{Kl}(M_S^c)$:

$$1 \xrightarrow{\sigma_*} S_0 \xrightarrow{\sigma_1} S_1 \xrightarrow{\sigma_2} S_2 \xrightarrow{\sigma_3} S_3 \xrightarrow{\sigma_4} \cdots$$

- Where $S_n \subseteq F^n(X)$ and choices exist in the game:

$$\begin{array}{ccc} S_n & \xrightarrow{\overline{\sigma_n}} & S_{n+1} \\ \downarrow & \sqsupseteq & \downarrow \\ F^n(X) & \xrightarrow[F^n(\gamma)]{} & F^{n+1}(X) \end{array}$$

- The *finite strategy outcome* $\text{out}_\sigma : 1 \rightarrow \widetilde{PM}_S^c(A^*B)$ is defined using the limit A^*B :



The finite linear-time semantics of games arise from the finitely completing strategies

Lemma 1

$$\gamma_n(x) = \{\sigma_{n-1} \circ \dots \circ \sigma_*(*) \mid \sigma \text{ is a strategy on } \gamma \text{ from } x\}$$

- This fails for $\mathcal{M}$ and $PM_{\mathbb{N}}$.

Theorem 2

$$\text{ftrace}_{\gamma}(x) = \bigcup_{\sigma \text{ is a strategy on } \gamma \text{ from } x} \text{out}_{\sigma}(*).$$

- Note: if σ does not enforce a subset of completed plays (a subset of A^*B) then $\text{out}_{\sigma}(*) = \emptyset$.

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Infinite Linear-Time Semantics

- Finite linear-time semantics arise from finitely completing strategies.
- Infinite linear-time semantics arise from ~~finitely completing~~ strategies.
- We stick with the monad $\widetilde{PP}$.
 - Homsets in $\mathbf{Kl}(\widetilde{PP})$ are complete lattices, composition is monotone in both arguments.
 - We have a complete lattice of homomorphisms:

$$\begin{array}{ccc}
 X & \xrightarrow{f} & A^\omega + A^*B \\
 \gamma \Downarrow & & \Downarrow \zeta \\
 A \times X + B & \xrightarrow[A \times f + B]{} & A \times (A^\omega + A^*B) + B
 \end{array}$$

- The greatest homomorphism gives us too much



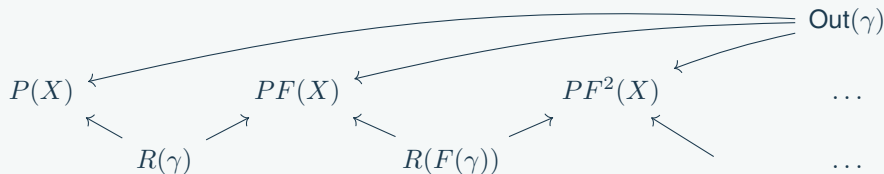
- If we record state information in the behaviour (moving from traces to executions), we can mend the homomorphism approach in two ways:
 - See it as being as completing a least fixed point
 - Or as a greatest fixed point (by restricting the lattice)
- What is the canonical definition of infinite linear-time semantics?

Infinite Linear-Time Semantics of Games as a Categorical Limit

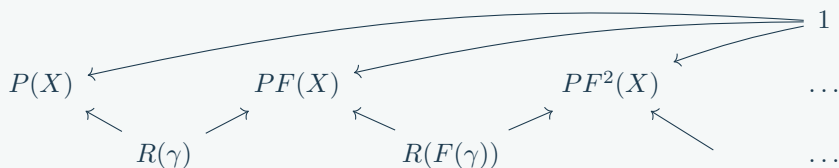
$R : \mathbf{Kl}(\widetilde{PP}) \rightarrow \mathbf{Rel}$ maps $f : X \rightrightarrows Y$ to $R(f) : P(X) \rightarrow P(Y)$:

$$R(f) = \{(U, \bigcup_{x \in U} V_x) \mid V_x \in f(x) \text{ for all } x \in X\}$$

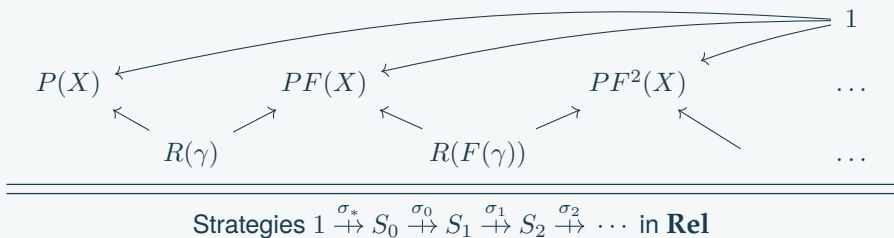
Viewing relations as spans in **Set**, we obtain:



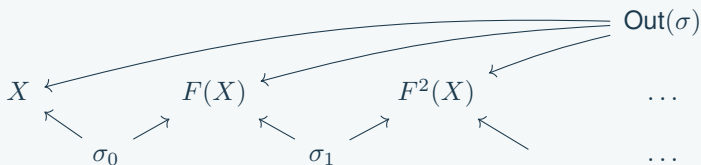
Define strategies as cones from 1:



Chain-Cone Correspondence



We can use limits to define infinite outcomes of strategies:



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Wrapping Up

1 Finite linear-time:

- We obtain a canonical semantics homomorphism for monads $\widetilde{PM}_S^c$, $\mathcal{M}_f$, and $PM_{\mathbb{N}}$.
- Proving the corresponding semantics arises from finitely completing strategies has been done for $\widetilde{PM}_S^c$.

2 Infinite linear-time:

- Using $\widetilde{PP}$, we can phrase semantics as a greatest fixed point, and as a completion of a least fixed point.
- Limits give a conceptual definition.

3 Future work:

- Incorporate quantitative environments in the infinite case.
- Is there a monad to handle *simple stochastic games*?
- Are there categories where infinite linear-time semantics arise canonically?
- Product construction and general synthesis algorithms.

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